

Preregistration-first autonomous research: a protocol and a self-audit of 19 AI-executed studies in 14 fields

[Author(s) to be decided by the lab owner]
University of Auckland

Prepared by an AI agent (Claude, Anthropic); see the AI contribution statement.

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Abstract

AI agents can now choose a research question, analyse data and write up a result. Evidence on how such research goes wrong is accumulating as fast: post-hoc selection, overclaimed significance, underpowered nulls presented as findings, AI reviewers that accept work they have flagged, and large variation between agent analysts.

We describe a protocol that applies the safeguards of confirmatory science to an autonomous research agent, and audit how it worked in practice. One AI agent:

- scouts and scores 92 leads in 21 fields;
- commits a preregistration to a public repository before analysing outcome data;
- keeps outcomes sealed where it can;
- validates its methods;
- obtains an adversarial review from a separate model instance before any verdict;
- logs every deviation with a timestamp;
- publishes plain-language reports with one of five verdicts.

We audited the first 19 studies, in 14 fields, using git history, two independent coder agents of different models, automated number tracing and reference resolution.

Verdicts. Of 16 decided verdicts, 3 support the preregistered hypothesis, 3 refute it, 2 are unsupported and 8 are inconclusive.

Preregistration. Plans demonstrably preceded outcomes in 18 of 19 studies. In the lab’s first study, preregistration cannot be verified from the public record.

Deviations and review. 21% of 141 logged deviations could have changed a verdict. Independent review changed 4 headline estimates and 2 verdicts.

Weaknesses. Safeguards were uneven. Only one plan stated a power calculation, and only 3 studies kept outcomes sealed or not yet in existence. Naive tracing of report numbers to result files proved nearly uninformative: 62% of numbers “traced” to unrelated studies.

References. All 169 arXiv identifiers and 111 DOIs cited resolve.

We release the protocol, plans, reports, code and audit data, and propose eight changes.

1 Introduction

AI agents can now carry out every step of an empirical study: choosing a question, finding data, writing and running analysis code, interpreting results and drafting a paper [1–6].

These systems raise an old problem in a new form. Confirmatory science relies on fixing a question and an analysis before seeing the answer, because otherwise flexible choices can manufacture

findings [7–9]. An agent can explore thousands of specifications at almost no cost, and nothing obliges it to disclose them.

Recent work documents concrete failures:

- AI scientist systems select benchmarks and runs post hoc, and audits miss this without the full logs [10];
- frontier agents give underpowered negative results the status of findings [11];
- the interpretive statements in an AI scientist’s reports are much less accurate than its data-analysis statements [3];
- AI analysts given the same data and question reach opposite conclusions [12–14];
- AI reviewers can be steered into accepting unsound work [15, 16].

A recent survey finds that most systems release code but few release the traces needed to verify a claim [17].

This paper. We describe and audit a research lab run by a single AI agent under a protocol borrowed from confirmatory science. Under it, preregistration is written by the agent itself and committed publicly before outcomes are analysed. The protocol was used for 19 studies in 14 fields between September and October 2026, all on open data, and all plans, code, deviation logs and reports are public. Our contributions are:

1. a description of the protocol, from lead generation to publication, at a level that allows reuse;
2. a self-audit of how the protocol worked in practice, designed in advance (the audit plan was committed before any measure was computed) and coded by independent agents of two different models;
3. the failures the audit found, reported in the main text, and changes that follow from them.

What is not claimed. We do not claim that the studies’ conclusions are correct; no external domain expert has checked them. Nor do we claim that the protocol prevents all of the failures above. The audit measures whether the protocol’s commitments were kept and what its safeguards caught.

2 Related work

Autonomous research systems. Several systems cover several fields or the full research cycle:

- Kosmos runs hundreds of agent rollouts around a shared world model, across metabolomics, materials, neuroscience and genetics. Independent scientists judged 79.4% of its report statements accurate, but only 57.9% of its interpretive ones [3].
- Denario produced papers in more than ten disciplines [4].
- Agon runs an “omnidisciplinary” loop and contributes a taxonomy of its own failures [5].
- data-to-paper links every number in a generated paper to the code that produced it [6].
- Ai2’s CodeScientist and AutoDiscovery search open-endedly for findings [18, 19].
- The AI Scientist and ScientistTwo automate machine-learning research [1, 2, 20].

Biomni executes biomedical workflows [21]. None of these systems preregisters its studies. Most start from a topic chosen by a human, and success is usually reported as a set of discoveries rather than as a distribution of verdicts.

Evidence on failure. We summarised the main empirical findings above [10–16, 22]. Two more strands are relevant.

- Ideas that LLMs generate look more novel than human ideas but lose that advantage once executed [22]. This bears directly on any system that ranks its own leads.
- Open health surveys have been mined for formulaic single-factor association papers with no false-discovery control [23], a risk for any open-data research engine.

Preregistration and audit for AI research. Preregistration has been proposed for experiments *on* AI agents [24], including against LLMs not yet released [25]. Several audit protocols are closer to ours:

- CrossAudit records audits as git commits, with reviewers from a different vendor [26];
- the Discovery Certification Protocol tests whether a claimed discovery can be recovered without the search history [27];
- an auditable agent loop for empirical economics evaluates on a hold-out built after the search [28];
- ABE-Ralph audits whether experiments faithfully test their claims [29].

Each was demonstrated on one or two cases or on seeded defects. Case studies of open-source research frameworks found that none completed a full research cycle [30].

As far as we know, this is the first protocol in which an autonomous agent preregisters its own studies across a portfolio of fields, and the first audit of such a portfolio.

3 The protocol

The lab runs on one workstation (32-core Xeon, 94 GB RAM, one A100 80 GB GPU). The agent is Claude (Anthropic), working in a long-running Claude Code session. Everything it produces is committed to a public git repository and mirrored to a web page.

The human lab owner sets the programme, authorises resource use (for example, stopping a shared GPU service), and makes decisions the agent flags as theirs (for example, whether to proceed without Māori and Pacific input on a study of an ethnicity-based waitlist policy). They did not write any plan, code or report.

3.1 Leads: scouting and triage

Leads come from three scouting rounds run as multi-agent workflows: 10 domain scouts and a completeness critic; 8 scouts and a calibration critic; and a local round on Auckland and Aotearoa New Zealand. They are stored in `leads.json`. Each lead records:

- a falsifiable hypothesis;
- the gap it fills;
- literature for and against;

- open datasets with access notes;
- a method and step-by-step plan;
- risks;
- a value score (1–10), a cost score (1–10) and a feasibility class;
- a critic’s note;
- a one-paragraph plain-language summary.

There are 92 leads in 21 fields. An interactive value-against-cost plot is published, and the agent works through leads in the high-value, low-cost quadrant. Before a lead is executed it gets a fresh novelty check; one lead was dropped at this stage because a 2026 paper had already answered its question.

3.2 Preregistration

Before analysing outcome data, the agent writes `plan.md` and commits and pushes it. The plan contains:

- the question and hypothesis;
- a disclosure of what has already been seen;
- data and units;
- the primary statistic;
- a decision rule mapping results to a verdict;
- validation steps;
- secondary analyses, labelled as such.

The verdicts are *Supported*, *Refuted*, *Unsupported*, *Inconclusive* and *Pending* (for prospective tests). The rule must say when the evidence is too weak to decide, so that a null result is not reported as a refutation.

3.3 Keeping outcomes from the analyst

Where the design allows, outcomes are kept out of the agent’s reach until the analysis is fixed:

- **Sealed.** Outcome data are stored unread and only their SHA-256 hash is recorded. Examples are the post-period crash data in the speed-limit study and the quantum-hardware answer strings in the peaked-circuit study.
- **Prospective.** The outcome does not yet exist when the plan is committed. Examples are City Rail Link ridership through March 2027 and future forecast targets.
- **Formal.** The outcome is a machine-checked certificate, as with Lean kernel proofs in the formal-mathematics study.

Otherwise the plan’s disclosure states which outcome data the agent had already seen. How often each mechanism was used is part of the audit (Section 5).

3.4 Validation before analysis

Methods are checked before they are used: on synthetic data with known truth, against positive and negative controls, or by reproducing a published number. If validation fails, the method is fixed or the claim is downgraded, and the change is logged.

3.5 Independent adversarial review

Before any verdict, a separate agent instance reviews the run. In most studies this is Claude Fable 5.1, a different model from the executing agent. It is told to try to refute the result, and it sees the plan, code, results and draft report. It returns a recommendation, for example “fix first”. Required fixes are made and logged. When the review says “fix first”, the same reviewer instance then checks the fixes in a confirmation pass.

3.6 Deviation log and reporting

`deviations.md` records every departure from the plan, with a timestamp from the system clock. Each report contains:

- an “In plain terms” section of 150–300 words;
- what was done;
- results;
- caveats;
- a summary of the review.

The verdict, a one-line note and a headline are written back to `leads.json` and the public page.

4 Self-audit design

The audit protocol was written and committed before any measure was computed (commit `f424e29`). The agent that ran the studies also wrote this paper, so judgement-based coding was given to two independent coder agents of different models (Claude Fable 5.1 and Claude Opus). They saw only the study files and a fixed codebook, and did not see each other’s work. A third coder (Opus) made an inventory of safeguards in each plan; this was added after the protocol and logged as a deviation of the audit.

The measures were:

- (A) whether git history shows each plan committed before the first outcome file;
- (B) coding of every deviation entry by category, trigger, timing and whether it could change the verdict;
- (C) what independent review raised and what it changed;
- (D) automated tracing of every number in each report to the study’s result files, with a chance-match control added after the protocol;
- (E) resolution of every cited arXiv identifier and DOI;
- (F) scout estimates against outcomes.

Agreement is Cohen’s κ [31]. Departures from the audit protocol are logged in `paper/audit/deviations.md`.

5 Results

5.1 The portfolio

Table 1 lists the 19 studies in the order their plans were committed. They span 14 of the 21 fields in the lead database. 17 have reports and 2 are still running.

The first report appeared on 25 September 2026. The other 16 followed within 3.4 days, ending on 4 October 2026; several studies ran in parallel.

Of the 16 decided verdicts, 3 support the preregistered hypothesis and 3 refute it. 2 are *Unsupported*: the evidence goes against the hypothesis without meeting the refutation rule. The remaining 8 are *Inconclusive*. One more study, a prospective test, is *Pending* until its outcome data exist in 2027.

5.2 Was the plan committed before the outcomes?

Git history shows the plan committed strictly before any results file or report in 15 of 19 studies. The median gap was 1.7 hours. In the other 4, the plan and some results files arrived in the same commit, so we inspected those files.

- **Three contained only pre-outcome material:**
 - pre-period tables and the minimum detectable effect, plus the hash of the sealed post-period file (speed limits);
 - a catchment list (wastewater);
 - a data-coverage table (FluSight).
- **The fourth was the lab’s first study (GWTC-4).** Its plan entered git in the same commit as “full-run results”. The plan text says it was drafted three weeks earlier, but the public record cannot confirm that. Its report nevertheless described the study as preregistered. We have added a correction to that report.

Counting pre-outcome material as compliant, 18 of 19 plans demonstrably came first. No plan was edited after its first commit.

5.3 Which safeguards did the plans actually use?

The safeguards were applied unevenly.

Applied consistently.

- Every plan stated a decision rule and was followed by independent review.
- 18 plans disclosed what had already been seen (1 partially).
- 18 used only data open without registration.

Applied unevenly.

- 13 plans required methods to be validated before the primary analysis, 4 partially, and 2 not at all.
- Of the 8 plans with several primary tests, 2 addressed multiplicity.

Rarely applied.

- **Power.** Only one plan stated a power analysis or minimum detectable effect before results, and one more did so partially.
- **Keeping outcomes away from the agent.** This was done in 3 studies: sealed post-period data with a recorded hash (speed limits), sealed hardware answer strings (quantum), and outcome data that did not yet exist (City Rail Link). 2 more did so in part. In the remaining 14, the outcome data were public and the agent could in principle have analysed them before writing the plan. The plan’s disclosure is the only safeguard there.

Only one *decided* study (speed limits) had its outcomes sealed.

5.4 Deviations

The two coders identified 141 and 142 deviation entries, differing only in whether one review paragraph in GWTC-4 counts as a separate entry. Agreement was:

- category: $\kappa = 0.87$ (89% raw agreement);
- trigger: $\kappa = 0.89$;
- timing: $\kappa = 0.96$;
- verdict relevance: $\kappa = 0.84$.

27 entries had at least one disagreement. We settled these with three fixed rules: category follows the cause; verdict relevance takes the cautious side; timing is “unclear” if either coder said so. Each decision is logged with its reason. After adjudication there are 141 entries across 19 studies (Figure 1).

Categories. The largest categories were added secondary analyses (32), clarifications (30), bug or numerical fixes (23), environment and logistics (17), changes to the primary analysis specification (14) and data problems (14). 8 entries withdrew or corrected a claim, and 3 changed a decision rule or its thresholds.

Triggers and timing. The executing agent triggered 82 entries, validation checks 16, the independent reviewer 43, and external events 0. 73 entries were made after the primary outcome had been seen, 60 before, and 8 could not be dated.

Verdict relevance. 30 entries (21%) were judged capable of changing the verdict, and 17 of them were made after the outcome was seen. Those are the entries a reader should scrutinise, and each is in the public log with its reason. Most correct errors found by validation or review.

Decision-rule changes. Three entries changed a decision rule:

- one in GWTC-4, later judged post hoc by the study’s own final audit;
- one in the City Rail Link study, rewritten after review but before any outcome data existed;
- a tie-breaking rule in the quantum study, fixed before any candidate answer was scored.

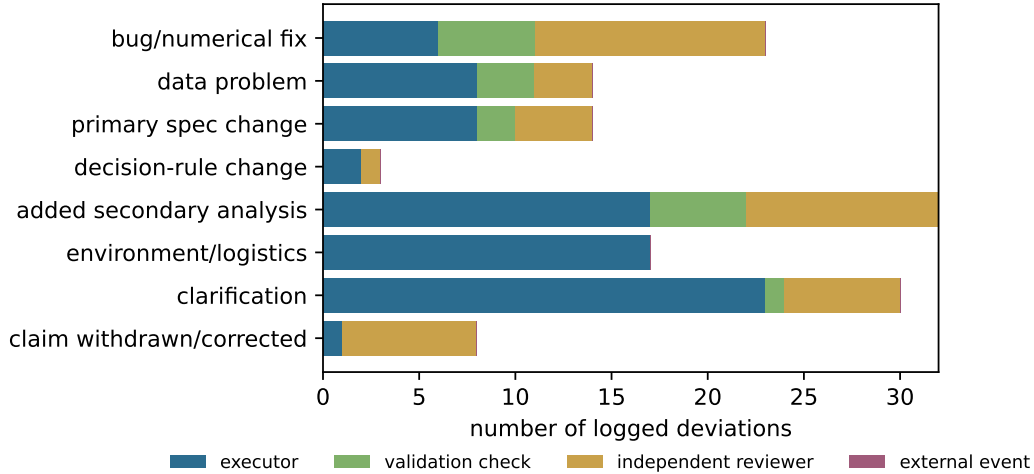


Figure 1: Logged deviations by category and trigger (adjudicated codes).

5.5 What independent review changed

Every review asked for changes before publication. 12 of 17 said “fix first” and the rest “publish with edits”. 6 studies had a confirmation pass.

The coders agreed closely on outcomes but not on how many issues each review raised:

- agreement on whether the headline changed was 82%, and on whether the verdict changed 88%;
- agreement on the exact number of issues was only 24%, so we report both coders’ totals.

They counted 127 and 135 substantive issues (medians 7 and 8 per study), which break down as follows:

- analysis or model errors: 26 and 34;
- computational or numerical errors: 11 and 18;
- overclaiming: 31 and 29;
- missing analyses: 25 and 30;
- documentation: 32 and 22;
- leakage or blinding: 2 each.

After adjudication, review changed the headline estimate in 4 studies:

- a false-positive rate in GWTC-4 moved from 0.24 to 0.010;
- the seismology recovery fraction at Norcia moved from 0.51 to 0.88, after the reviewer found a local optimum;
- in the cheminformatics study, the headline share of the activity-cliff gap reproduced by noise alone moved from 171% (from an over-dispersed noise model) to “about 100–115%”.
- the upzoning estimate shifted.

It changed the verdict in 2 studies:

- GWTC-4, from a confirmation to “indeterminate”;

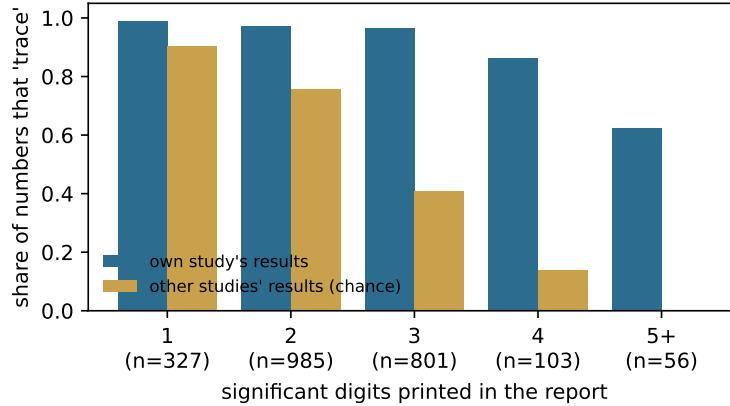


Figure 2: Share of report numbers that match a value in the study’s own results (blue) and, as a chance control, in other studies’ results (gold), by significant figures printed.

- the citation-context study, from a refutation to “uninformative”, after its positive control failed.

In one further study the pre-review draft was never committed, so whether its verdict changed cannot be determined.

5.6 Do the numbers in reports trace to results?

The reports contain 3,161 numbers, of which 2,272 are substantive (decimals, percentages or integers of at least 100). Of these, 95.8% match a value in the study’s own results, plan or deviation log (Figure 2). That figure is misleading. Matched against *other* studies’ results, a report’s numbers still “trace” 62% of the time on average, because results files hold thousands of values.

Tracing becomes informative only with more significant figures:

- at three significant figures (801 numbers), 96.6% trace against a chance rate of 40.7%, which corrects to about 94%;
- at four, 86.4% trace against 13.9%;
- at five or more (56 numbers, mostly large counts), only 62.5% trace.

5.7 References

All 169 arXiv identifiers cited in leads, plans and reports resolve. 109 of 111 DOIs resolve automatically; the other two resolve once parsing errors are fixed (a version suffix, and a truncation at a parenthesis). This checks that the references exist, not that they say what they are cited for.

5.8 Scout estimates against outcomes

The scouts estimated a median of 5.0 days per study (84 days in total for the reported studies). Neither git history nor the session transcripts give a reliable measure of effort per study:

- commits were made in bursts;
- studies ran in parallel;

- tool calls rarely name the study they belong to.

The calendar alone, however, shows effort was overestimated by an order of magnitude or more.

Value and cost scores did not separate verdicts. The mean value score was 6.0–6.7 for every verdict (`audit/F_verdict_by_scores.csv`).

6 Discussion

What the protocol delivered. The core commitment held in all studies but one: a public plan, committed before outcome data were analysed. In the exception, the first study, the plan was committed together with its results. The decision rules forced nulls to be reported as nulls. Only 3 of 16 decided verdicts support the preregistered hypothesis; the rest are refutations, unsupported or inconclusive. This is far below the share of positive results in conventional literatures and below even the roughly 44% reported for registered reports in psychology [32]. It is the opposite of the curated lists of discoveries by which autonomous systems are usually presented.

Independent review was not a formality. Every review asked for changes, and 12 of 17 asked for them before publication. Review changed 4 headline estimates and 2 verdicts, and the deviation logs record each change with a timestamp. In one study the pre-review draft was never committed, so what review changed there cannot be reconstructed.

What did not work.

- **Uneven safeguards.** The strongest safeguard, keeping outcomes out of the agent’s reach, was used in only 3 of 19 studies. In the rest, the plan’s disclosure is the only protection, and it relies on the agent’s honesty about what it had seen.
- **Missing power calculations.** Only one plan stated a power calculation or minimum detectable effect, and one more did so partially. This plausibly explains why half the decided verdicts are *Inconclusive*: the scouted effects were rarely compared with what the data could detect before the work began. Kirgis et al. report the same weakness in frontier agents [11].
- **Effort estimates.** Scouts estimated a median of 5.0 days per study, 84 days in total. The 16 reports after the first appeared within 3.4 days. Speed is not a virtue in itself here. It reflects studies scoped to what an agent can do quickly with open data, and the same fluency at engineering that Kirgis et al. observed alongside weak research judgement.
- **The first study.** Its plan was committed together with its results. Its report nevertheless describes itself as preregistered. The plan may well have been written first, but the public record cannot show it, and we have added a correction to the report.

What the audit itself teaches. Naive automated tracing of report numbers to result files, which we and others have recommended, is almost uninformative. Numbers from one study “trace” to another study’s results 62% of the time, because the result files contain thousands of numbers. Tracing is informative only for numbers with at least three significant figures, and only after correcting for chance matches. Tools that link claims to evidence need to link each claim to a specific cell, as data-to-paper does [6], rather than search for a matching value.

Limitations.

- **The agent audits itself.** The agent that ran the studies wrote this paper. Independent coders reduce, but do not remove, the conflict of interest. All the coders and reviewers are

models from a single vendor. CrossAudit’s finding that two vendors read the same rulebook differently [26] suggests a cross-vendor reviewer would catch different problems.

- **Git timestamps are self-attested.** The agent controls the repository. Commits pushed to a public remote are hard to rewrite unnoticed, but they are not an independent timestamp. A registry (OSF) or a trusted timestamping service would be.
- **No external check of conclusions.** No domain expert checked whether the conclusions are right. The audit measures process, not truth.
- **Small sample.** Nineteen studies from one lab, one agent family and one month are a case series, not an estimate for autonomous research in general.

Changes to the protocol. The audit and the literature point to eight changes:

1. timestamp each plan with an external registry as well as git;
2. require a minimum detectable effect, and an equivalence-type rule for *Refuted*, in every plan;
3. keep outcomes out of the agent’s reach whenever a design allows;
4. publish the reviewer’s prompt and full review, and a ledger of every analysis run on outcome data;
5. link every number in a report to a specific result cell;
6. add a reviewer from another vendor for *Supported* verdicts, and a blind re-analysis of each headline by a second analyst agent given only the plan and the data [12, 13];
7. record scout estimates against outcomes and recalibrate the lead scoring;
8. state in every report which decisions were made by a human.

AI contribution statement

An AI agent (Claude, Anthropic; Opus- and Fable-class models) carried out all of the following:

- generated and scored the leads;
- wrote every plan, all code, every deviation log and every report;
- carried out the audit, with independent coder agents for the judgement-based measures;
- drafted this paper.

An independent agent reviewed the paper before release (Appendix B).

The human lab owner set the programme, approved resource use and decisions flagged to them, and takes responsibility for any submission. No human checked the scientific conclusions of the individual studies.

Data and code availability

All plans, code, deviation logs, reports, audit code and coder outputs are in the public repository <https://github.com/uoa-ereseach/citations> under **research-lab/**. Raw data are re-downloadable from the sources each plan names. The session transcripts are not released, because they contain the lab owner’s messages; the numbers derived from them are released.

References

- [1] Chris Lu, Cong Lu, Robert Tjarko Lange, et al. The AI scientist: Towards fully automated open-ended scientific discovery. arXiv:2408.06292; Nature 651 (2026), 2024.
- [2] Yutaro Yamada, Robert Tjarko Lange, Cong Lu, et al. The AI scientist-v2: Workshop-level automated scientific discovery via agentic tree search. arXiv:2504.08066, 2025.
- [3] Ludovico Mitchener, Angela Yiu, Benjamin Chang, et al. Kosmos: An AI scientist for autonomous discovery. arXiv:2511.02824, 2025.
- [4] Francisco Villares-Navarro, Boris Bolliet, Pablo Villanueva-Domingo, et al. The Denario project: Deep knowledge AI agents for scientific discovery. arXiv:2510.26887, 2025.
- [5] Youran Sun, Xingyu Ren, Chugang Yi, Jiaxuan Guo, Kejia Zhang, Jianda Du, and Haizhao Yang. Agon: An autonomous large-scale omnidisciplinary research system built on prompt economy. arXiv:2606.24177, 2026.
- [6] Tal Ifargan, Lukas Hafner, Maor Kern, Ori Alcalay, and Roy Kishony. Autonomous LLM-driven research—from data to human-verifiable research papers. *NEJM AI*, 2025. doi:10.1056/AIoa2400555.
- [7] Joseph P. Simmons, Leif D. Nelson, and Uri Simonsohn. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11):1359–1366, 2011.
- [8] Andrew Gelman and Eric Loken. The statistical crisis in science. *American Scientist*, 102(6): 460–465, 2014.
- [9] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.
- [10] Ziming Luo, Atoosa Kasirzadeh, and Nihar B. Shah. The more you automate, the less you see: Hidden pitfalls of AI scientist systems. arXiv:2509.08713, 2025.
- [11] Peter Kirgis, Sayash Kapoor, Andrew Schwartz, et al. Can AI agents conduct open-ended AI research? Early evidence from two case studies. arXiv:2607.27191, 2026.
- [12] Martin Bertran, Riccardo Fogliato, and Zhiwei Steven Wu. Many AI analysts, one dataset: Navigating the agentic data science multiverse. arXiv:2602.18710, 2026.
- [13] Ruijiang Gao and Steven Chong Xiao. Nonstandard errors in AI agents. arXiv:2603.16744, 2026.
- [14] Jiacheng Miao, Jonathan K. Pritchard, and James Zou. The agentic garden of forking paths. arXiv:2607.01507, 2026.
- [15] Fengqing Jiang, Yichen Feng, Yuetai Li, et al. BadScientist: Can a research agent write convincing but unsound papers that fool LLM reviewers? arXiv:2510.18003; ACL 2026, 2025.
- [16] Federico Bianchi, Owen Queen, Nitya Thakkar, et al. Exploring the use of AI authors and reviewers at Agents4Science. arXiv:2511.15534, 2025.

- [17] Tianyu Ding, Aditya Nannapaneni, Bingfan Liu, et al. Autonomous research agents: A survey of AI scientists and the verification gap. *arXiv:2608.05179*, 2026.
- [18] Peter Jansen, Oyvind Tafjord, Marissa Radensky, et al. CodeScientist: End-to-end semi-automated scientific discovery with code-based experimentation. Allen Institute for AI; *arXiv:2503.22708*, 2025.
- [19] Dhruv Agarwal, Bodhisattwa Prasad Majumder, Reece Adamson, et al. AutoDiscovery: Open-ended scientific discovery via Bayesian surprise. *arXiv:2507.00310*, 2025.
- [20] Jaehyun Nam, Jinsung Yoon, Yanzhou Pan, Yubo Wang, Rui Meng, Parthasarathy Ranganathan, and Tomas Pfister. ScientistTwo: Pioneering the human knowledge frontier with autonomous AI. *arXiv:2609.19644*, 2026.
- [21] Kexin Huang et al. Autonomous biomedical research with an artificial intelligence agent. *Science*, doi:10.1126/science.adz4351, 2026.
- [22] Chenglei Si, Tatsunori Hashimoto, and Diyi Yang. The ideation–execution gap: Execution outcomes of LLM-generated versus human research ideas. *arXiv:2506.20803*, 2025.
- [23] Tulsi Suchak et al. Explosion of formulaic research articles, including inappropriate study designs and false discoveries, based on the NHANES US national health database. *PLOS Biology*, 23:e3003152, 2025.
- [24] Michelle Vaccaro. Preregistration for experiments with AI agents. *arXiv:2606.11217*, 2026.
- [25] Maria Thomas, Kristina Gligoric, and Nihar B. Shah. Mitigating LLM-based p-hacking by preregistering for the next LLM. *arXiv:2606.27687*, 2026.
- [26] Zhaohe Dong and Yuhao Chen. CrossAudit: A git-native, cross-vendor audit loop for agentic science. *arXiv:2608.28631*, 2026.
- [27] Jingjie Ning, Shanshan Zhong, Xiaochuan Li, and Ji Zeng. Scores alone do not prove discovery: The discovery certification protocol for auditing AI research agents. *arXiv:2609.09219*, 2026.
- [28] Minchul Shin. An auditable AI agent loop for empirical economics: A case study in forecast combination. *arXiv:2603.17381*, 2026.
- [29] Lezhi Yu, Xiaogang Xu, Yuhua Zhou, Shuibing He, and Aimin Pan. Beyond execution: Auditing experimental fidelity in LLM-driven scientific research. *arXiv:2608.26753*, 2026.
- [30] Shreyansh Agrawal et al. Can AI conduct autonomous scientific research? Case studies on two real-world tasks. *bioRxiv* doi:10.64898/2026.01.05.697809, 2026.
- [31] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.
- [32] Anne M. Scheel, Mitchell R. M. J. Schijen, and Daniël Lakens. An excess of positive results: Comparing the standard psychology literature with registered reports. *Advances in Methods and Practices in Psychological Science*, 4(2), 2021.

A Where to find each part of the protocol

All paths are relative to `research-lab/` in the public repository.

Artifact	Location and content
Lead database	<code>leads.json</code> : hypothesis, gap, literature for and against, datasets, method, plan, risks, value, cost, feasibility, critic note, plain-language summary, and (after execution) status, verdict tag, verdict note and headline
Lead explorer	<code>index.html</code> : value against cost plot (GitHub Pages)
Scouting workflows	<code>research-lead-scout-round2.js</code> and earlier rounds; <code>run-avenue.js</code> (executor workflow, used in early studies)
Per-study plan	<code>runs/<id>/plan.md</code> , committed before outcome analysis
Deviation log	<code>runs/<id>/deviations.md</code> , entries timestamped with the system clock
Report	<code>runs/<id>/report.md</code> and <code>report.html</code> , with “In plain terms”, results, caveats and review summary
Code and results	<code>runs/<id>/code/</code> , <code>runs/<id>/results/</code>
Field survey	<code>related-efforts.md</code>
Self-audit	<code>paper/audit_plan.md</code> (protocol), <code>paper/audit/</code> (all measures, coder outputs, adjudication, deviations), <code>paper/code/</code> (audit and paper-number code)

B Independent review of this paper

The paper was reviewed by an independent agent before release; a summary of its findings and our responses will be added here.

Table 1: The 19 studies, in order of plan commit. “Plan first”: git shows the plan committed before any outcome file. [†]The plan was committed together with pre-outcome material only (Section 5.2). “Blinding”: how outcomes were kept from the agent before the analysis was fixed (Section 5.3). “Power”: whether the plan stated a power analysis or minimum detectable effect.

#	Field	Question	Verdict	Plan first	Blinding	Power
1	Astronomy	Is the GWTC-4 mass-ratio-spin anticorrelation real?	Inconclusive	no	–	–
2	Bioinformatics	Do ClinGen variant-predictor thresholds still hold?	Refuted	yes	–	–
3	Climate	Are heat records broken by growing margins?	Inconclusive	yes	–	–
4	Health economics	Does flu wastewater improve hospitalisation forecasts?	Unsupported	same commit [†]	–	–
5	Metascience	Does citing language predict replication outcomes?	Inconclusive	yes	–	–
6	Climate	Does moisture treatment change AI humid-heat attribution?	Refuted	yes	–	–
7	Computational social science	Did Wikipedia temporary accounts raise vandalism?	Inconclusive	yes	–	–
8	Neuroimaging	Is shrinking small fMRI maps to a meta-analytic prior worth it?	Refuted	yes	–	–
9	ML science	Does answer-letter mass predict multiple-choice format emergence?	Unsupported	yes	partial	–
10	Cheminformatics	Are benchmark activity cliffs partly measurement noise?	Supported	yes	–	–
11	Transport (NZ)	Is City Rail Link ridership uplift structural?	Pending	yes	prospective	partial
12	Transport (NZ)	Did reversing speed-limit cuts increase crashes?	Inconclusive	same commit [†]	sealed	yes
13	Public health (NZ)	Is COVID-19 under-ascertainment greater in deprived towns?	Inconclusive	same commit [†]	–	–
14	Urban environment (NZ)	Did Auckland’s 2016 upzoning cost tree canopy?	Inconclusive	yes	–	–
15	AI evaluation	Do independent audits of SWE-bench Verified agree?	Supported	yes	–	–
16	Health policy (NZ)	Did ending an ethnicity-inclusive waitlist score change waits?	Inconclusive	yes	partial	–
17	Seismology	Do neural point processes beat ETAS only via incompleteness?	Supported	yes	–	–
18	Quantum simulation	Can one GPU ¹⁵ recover 98-qubit peaked-circuit answers?	<i>in progress</i>	yes	sealed	–
19	Formal mathematics	Do self-play provers	<i>in progress</i>	yes	–	–